

CHAPTER 5 : ACIDS, BASES AND SALTS

Topic:-

Obtain a sample of rainwater. Add a few drops of universal indicator. Measure its pH. Describe the nature of the sample and write its effect on the living world.

Introduction:-

Water is the universal solvent essential for all biochemical life on Earth. In its pure, distilled state, water is neutral with a pH value of exactly 7.0 at 25°C.

However, when rain falls from clouds through the atmosphere, it dissolves airborne gases. Normal, unpolluted rainwater naturally dissolves atmospheric carbon dioxide (CO₂), forming weak carbonic acid (H₂CO₃), which gives clean rainwater a slightly acidic pH of approximately 5.6 to 6.5.

In heavily industrialized cities and dense traffic regions, the burning of fossil fuels (coal, diesel, petrol) releases toxic gaseous pollutants—specifically sulphur dioxide (SO₂) and nitrogen oxides (NO_x). In the upper atmosphere, these oxides react with atmospheric moisture and oxygen to form strong sulphuric acid (H₂SO₄) and nitric acid (HNO₃). When this precipitation falls, it is termed **Acid Rain** (pH below 5.6).

In this project, we collected a fresh rainwater sample during an active rain shower in our locality, tested it with a universal indicator solution and standard pH color chart, determined its exact pH, characterized its chemical nature, and investigated its ecological impacts on living organisms.

❖ Key Highlights of Rainwater pH Testing:-

- Universal indicator provides a continuous spectrum of distinct colors from red (pH 1) to purple (pH 14).
- Unpolluted natural rain has a pH around 6.0; industrial acid rain can drop to pH 4.0 or lower.
- Acidic precipitation disrupts soil nutrient balance, damages aquatic ecosystems, and harms buildings.

Aim & Apparatus Setup:-

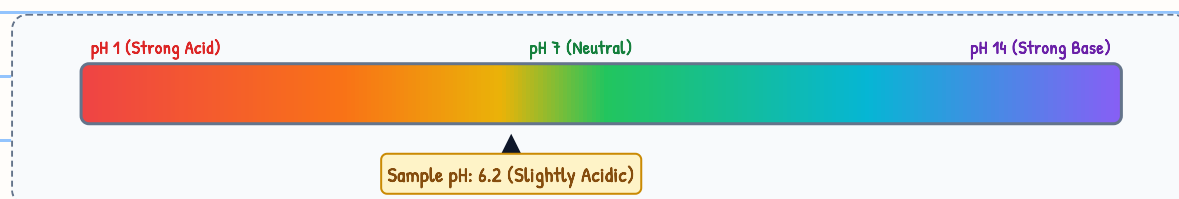
Aim: To collect a fresh rainwater sample, determine its pH value using universal indicator solution, describe its chemical nature, and analyze its physiological effects on the living world.

❖ Objectives of the Project:-

- To collect uncontaminated rainwater directly from open sky using acid-washed glassware.
- To determine the pH of rainwater and compare it with tap water, boiled water, and pond water.
- To interpret the color shift against the calibrated Sorenson universal pH color chart.
- To study the harmful biological consequences of acidic rainwater on plants, fish, and humans.

❖ Materials & Reagents Required:-

- Freshly collected rainwater sample (collected in clean glass container after 15 min of initial shower).
- Universal indicator solution with calibrated reference color comparison chart (pH 1 to 14).
- Clean borosilicate test tubes, test tube stand, glass dropper, and digital pH paper strips.
- Control samples: Municipal tap water, local freshwater pond water, and distilled water.



Experimental Procedure & Testing:-

The testing of rainwater pH was conducted in our school chemistry laboratory following standard experimental procedures:

❖ *Step-by-Step Testing Procedure:-*

- **Step 1 (Sample Collection):** Placed a thoroughly cleaned and rinsed 500 ml glass beaker on an elevated open school terrace away from tree canopies and building overhangs during active rain. Waited 15 minutes after rain started to avoid initial dust washout.
- **Step 2 (Sample Preparation):** Transferred 10 ml of the collected rainwater sample into a clean, dry borosilicate test tube. Labeled it Test Tube A.
- **Step 3 (Indicator Addition):** Using a clean glass dropper, added 4 to 5 drops of universal indicator solution to the rainwater sample. Swirled the test tube gently to mix uniformly.
- **Step 4 (Color Matching):** Placed the test tube against a clean white background in daylight and carefully compared the resulting liquid color against the standard pH reference chart.
- **Step 5 (Cross-Verification):** Repeated the measurement using narrow-range digital pH paper strips and recorded the observations alongside control samples of tap water and pond water.

Observations & Water Sample pH Table:-

Our comparative laboratory evaluation yielded the following experimental observations for rainwater alongside local environmental water samples:

Water Sample	Universal Indicator Color	Observed pH	Chemical Nature	Ecological Status
Rainwater Sample	Yellowish Green	6.2	Slightly Acidic	Normal natural range
Distilled Water	Pure Green	7.0	Neutral	Safe reference standard
Municipal Tap Water	Dark Greenish Cyan	7.4	Slightly Alkaline	Safe for drinking
Local Pond Water	Greenish Yellow	6.8	Near Neutral	Supports fish & algae
Simulated Acid Rain	Bright Reddish Orange	4.2	Strongly Acidic	Severely hazardous

The tested rainwater sample displayed a yellowish-green tint indicating a pH of 6.2, which falls within the healthy natural baseline for non-polluted suburban precipitation.

Scientific Analysis & Nature of Sample:-

Analyzing our experimental findings reveals important scientific truths regarding atmospheric chemistry and rainwater acidity:

❖ *Chemical Nature of the Sample:-*

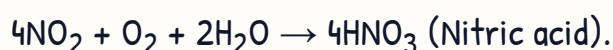
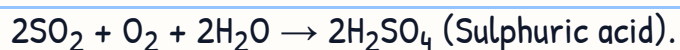
- **Natural Dissolution of Carbon Dioxide:** Rain falling through clean air naturally dissolves CO_2 , setting up the reversible equilibrium:



This naturally lowers rainwater pH from 7.0 to around 6.0 – 6.5 without causing environmental harm.

- **Absence of Severe Acid Rain:** Because our sample measured pH 6.2 (well above the critical threshold of 5.6), our locality does not currently suffer from acute acid rain. This reflects low industrial emissions of SO_2 and NO_2 in our immediate residential vicinity.

- **Chemical Mechanism of Dangerous Acid Rain:** In industrial centers, catalytic oxidation transforms emissions into strong mineral acids:



Effects of Acid Rain on the Living World:-

When rainwater pH drops significantly below 5.6, it inflicts severe multi-dimensional devastation across terrestrial and aquatic ecosystems:

❖ *Devastating Ecological Effects of Acidic Precipitation:-*

- **Damage to Plant Foliage & Crops:** Acid rain corrodes the protective waxy cuticle layer of leaves. This impairs stomatal function, damages chlorophyll, and severely reduces photosynthetic food production in trees and agricultural crops.
- **Leaching of Vital Soil Minerals:** High concentrations of H^+ ions leach away essential soil cations like calcium (Ca^{2+}), potassium (K^+), and magnesium (Mg^{2+}), leaving soils barren while liberating toxic aluminum ions (Al^{3+}) that stunt plant roots.
- **Destruction of Aquatic Life:** Lakes and ponds with pH below 5.0 cannot sustain freshwater fish, frog eggs, or mollusks. Acid dissolves fragile calcium carbonate shells of snails and impairs fish gills, leading to mass fish mortality.
- **Corrosion of Historic Monuments:** Acid rain reacts with marble and limestone ($CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$), causing "marble cancer" as seen on the Taj Mahal.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Mastered laboratory techniques of measuring liquid pH accurately using universal indicator and color charts.
- Learned the chemical difference between natural slightly acidic rain (pH ~6.0) and hazardous acid rain (pH <5.6).
- Recognized how air pollution directly alters rain chemistry and disrupts fragile living ecosystems.

❖ Project Conclusion:-

Our collected rainwater sample registered a pH of 6.2, characterizing it as slightly acidic due to natural atmospheric carbon dioxide dissolution, and safe for the local biosphere. However, preserving our clean rainwater demands strict vigilance, reduction of vehicular fossil fuel combustion, transition to solar and renewable energy, and extensive tree plantation across urban belts.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 5: Acids, Bases and Salts).
- Central Pollution Control Board (CPCB), Government of India – Annual Report on Rainwater Quality Monitoring.
- United States Environmental Protection Agency (US EPA) – Educational Resources on Acid Rain and the pH Scale.